

Tugas 7 Termodinamika

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1. Forward Difference ($h = 20\text{ s}$)

Dik: suhu benda : 10 s

Dit:

$t(\text{s})$	0	10	20	30	40
$T(t)(^{\circ}\text{C})$	85,0	76,2	68,5	61,9	56,1

jawab:

$$\begin{aligned}\left. \frac{dT}{dt} \right|_{t=t_i} &= \frac{T(t_i + h) - T(t_i)}{h} \\ &= \frac{T(20 + 10) - T(20)}{10} \\ &= \frac{T(30) - T(20)}{10} \\ &= \frac{61,9 - 68,5}{10} \\ &= \frac{-6,6}{10} = -0,66^{\circ}\text{C/s} \quad //\end{aligned}$$

2. Backward Difference ($h = 1,0\text{ s}$)

Dik: suhu = $0,12$, $C = 470\text{ nF} = 470 \times 10^{-6}\text{ F}$

Dit:

$t(\text{s})$	0,0	0,2	0,4	0,6	0,8	1,0
$V(t)(\text{V})$	5,00	3,70	2,74	2,103	1,50	1,11

jawab:

$$\begin{aligned}\left. \frac{dV}{dt} \right|_{t=1,0} &= \frac{V(t_i) - V(t_i - h)}{h} \\ &= \frac{V(1,0) - V(1,0 - 0,12)}{0,12} \\ &= \frac{V(1,0) - V(0,88)}{0,12} \\ &= \frac{1,11 - 1,50}{0,12} \\ &= \frac{-0,39}{0,12} = -1,95\text{ V/s} \quad //\end{aligned}$$

$$I(t) = C \frac{dV}{dt} = 470 \times 10^{-6} (-1,95)$$

$$= -9,115 \times 10^{-4}\text{ A} \approx -0,917\text{ mA} \quad //$$

3. Central Difference ($t = 1,5$)

Dik : suhu = 0,5

Data							
t (s)	0,0	0,5	1,0	1,5	2,0	2,5	3,0
x (t) (m)	0,00	0,62	1,48	2,98	3,92	5,90	7,32

Jawab :

$$\begin{aligned}
 \frac{dx}{dt} \Big|_{t=1,5} &= \frac{x(t+h) - x(t-h)}{2h} \\
 &= \frac{x(1,5 + 0,5) - x(1,5 - 0,5)}{2(0,5)} \\
 &= \frac{x(2,0) - x(1,0)}{1,0} \\
 &= \frac{3,92 - 1,48}{1,0} \\
 &= \frac{2,44}{1} = 2,44 \text{ m/s}
 \end{aligned}$$